

HAT-P-36b

R-1186

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_220520 · created 2026-07-31T05:10:38+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459719.7802 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1414 +/- 0.008
Transit depth (Rp/Rs)^2	2.0 +/- 0.23 [%]
Orbital Inclination (inc)	89.94 +/- 1.52
Airmass coefficient 1 (a1)	1.035 +/- 0.012
Airmass coefficient 2 (a2)	-0.0279 +/- 0.007
Scatter in the residuals of the lightcurve fit is	1.19 %
Best Comparison Star	#3 - [159, 306]
Optimal Aperture	3.87
Optimal Annulus	5.79
Transit Duration (day)	0.0988 +/- 0.001

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1414 $\pm$ 0.008 vs 0.126 (deviation 1.88 σ)
Duration fit vs published	0.0988 d vs 0.0925 d (deviation 6.16 σ)
Mid-time O-C	4.0 min
Depth SNR	17.3
Residual scatter	1.19 %

Light curve

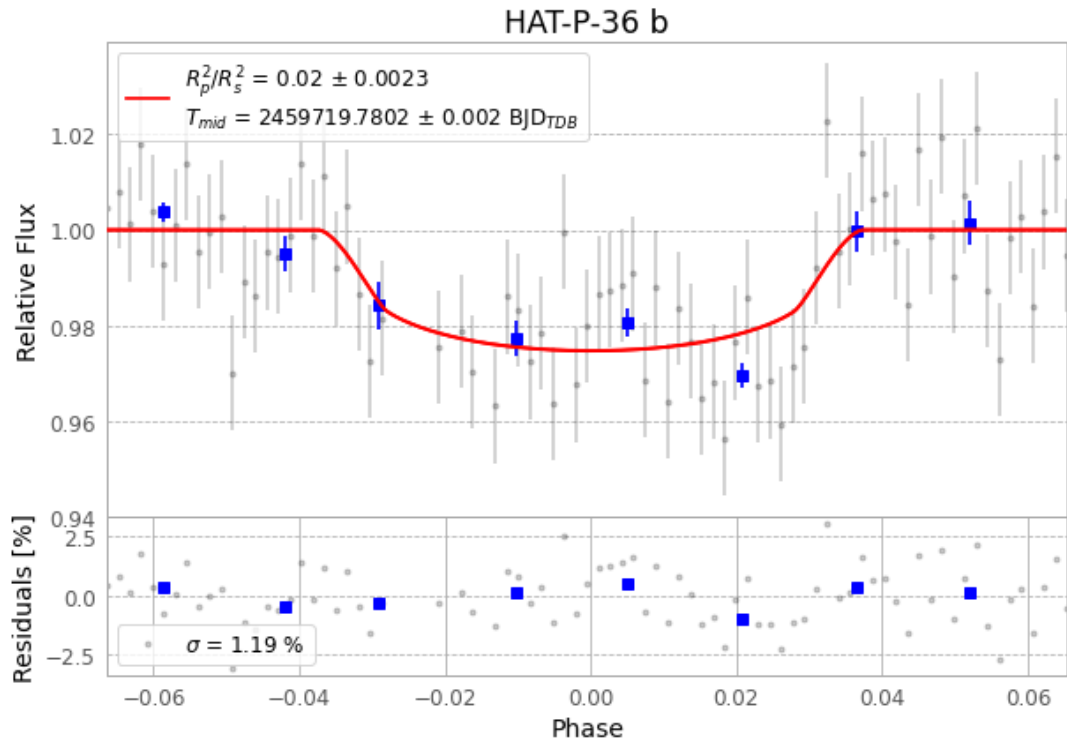


Figure 1 — FinalLightCurve_HAT-P-36 b_2022-05-19.png (SHA-256 257a36c2270a3889...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [303, 162], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 593dec2db359ea36...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

85 FITS frames (56 MB), HATP-36220520043313.FITS → HATP-36220520084512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-220520.log (b88c19ede209...), FinalLightCurve_HAT-P-36 b_2022-05-19.png (257a36c2270a...), FinalLightCurve_HAT-P-36 b_2022-05-19.csv (910399204bea...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 05:10Z.